Group 2
Vertical Injection Molding Machine

ME 3250L Section 4

Machine Design Laboratory

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Luke Esperiquette – Individual Contribution Report

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1. Abstract

In this project, our group designed and built a vertical injection molding machine. I contributed to both the organizational and technical aspects of the project. I organized meeting times, arranged access to lab spaces, and sourced materials at low or no cost. I played a major role in machining several key components and helped wire and troubleshoot the version 1 motor system. I also provided design input, including the idea to use a spring-loaded middle plate and to replace aluminum extrusion with threaded rods. Throughout the build, I helped solve real-time issues, kept the group accountable, and made sure we stayed on schedule.

2. Introduction

The goal of our project was to build a functioning vertical injection molding machine that could simulate basic thermoplastic molding processes. Our work required mechanical design, machining, electrical integration, and effective team coordination. I contributed across all of these areas, helping guide both the planning and execution phases.

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3. Design & Contribution

Team Coordination and Scheduling

I helped schedule and organize regular group meetings, including coordinating access to the Innovation Orchard Maker Space through communication with Mr. Martin White. Since the space was undergoing renovations, we had to book it by appointment only. I also helped set up an off-campus machining day in Rancho Cucamonga with Elijah, where we worked remotely and received troubleshooting support from Nico via phone.

One of the biggest challenges was scheduling meetings around everyone's busy lives. We often struggled with missed messages or forgotten plans. I took initiative to push the team to meet deadlines, follow through, and stay engaged throughout the semester.

Tool Access and Materials

I helped ensure that we had access to the necessary shop tools during build days. I also brought in materials for the project, many of them at discounted prices or for free. Which helped us stay under budget. This included sourcing basic structural hardware, fasteners, and electrical components.

Design Contributions

One of my key design contributions was suggesting the use of a spring-loaded middle plate instead of fabricating a second mid plate. This saved us significant build time and reduced material costs. The spring mechanism allowed the barrel to press down onto the mold during injection, then return to the raised position, freeing up the mold for access and reset.

Another major contribution was the recommendation to use threaded rods instead of aluminum extrusions for the machine's frame. While initially chosen to save costs, the threaded rods also gave us flexibility in adjusting the height of each machine level and simplified overall assembly.

When our initial motor setup couldn't drive the lead screw in both directions, I proposed using a high-torque backup motor system. We repurposed a 1000 ft-lb Milwaukee drill to provide bidirectional power. With help from Hudson, we discovered that by using two jam nuts to lock onto the lead screw, we could apply torque directly and consistently. I also designed the support

structure to secure the drill on the platform, ensuring it could deliver force safely without slipping or misalignment. This workaround was essential to restoring functionality to the system. Combined with the lever clamp design, which had a mechanical advantage of 2.75:1, the system achieved an estimated closing force of 2750 lb-ft. This was more than sufficient to securely compress the mold halves during injection, and it highlighted how we were able to use off-the-shelf tools in a creative, high-performance way. This can be seen in Figure 11.

Machining and Manufacturing

I helped machine the version 1 motor-to-lead screw adapter on the lathe and used the drill press to fabricate the clamp-to-mold adapter. I also helped James Willis machine parts for the clamp mechanism. One machining challenge was drilling large holes; going directly to a large bit caused vibration and instability, so I used progressively larger drill bits to reach the final size which can be seen in Figure 10. While threading holes, I had to be careful with tap speed to avoid damaging the tool and learned to apply oil during both tapping and drilling to improve quality and reduce wear which can be seen in Figure 5.

Electrical Work and Motor System Integration

I wired and tested the version 1 motor system, attempting to run a 3-phase motor using a 40V battery supply. The system powered the motor in one direction, but I couldn't get it to reverse, which led to the development of the backup drill system. I also designed the mounting interface between the motor and the frame, ensuring proper alignment and smooth torque transmission to the lead screw.

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4. Analysis

I performed a mix of informal testing and functional analysis throughout the build. For machined components, I used calipers to check tolerances and ensured tight fits between fasteners and mating parts. After turning the lead screw adapter on the lathe, I verified that its bore aligned concentrically with the shaft and that it transferred motion without visible play or wobble.

I tested the original motor setup using a multimeter to measure voltage output and verify electrical continuity across the wiring. I confirmed the motor's ability to spin under load in one direction, and diagnosed the directional issue as a limitation of the current wiring setup. As a result, I pivoted to developing the high-torque drill backup system.

In the drill setup, I confirmed that the locking nut assembly on the lead screw held under load and allowed effective torque transfer from the Milwaukee drill. I also checked alignment between the drill body and the platform to minimize off-axis motion. Real-world testing involved applying force through the drill and observing the consistent linear movement of the clamping system.

The spring plate system was tested manually to ensure it provided sufficient resistance to return the barrel to its raised position after an injection cycle. This was key to avoiding the need for additional actuators or mechanical lifts.

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5. Conclusion

This project gave me valuable experience in both machine design and real-world problem solving. I contributed heavily to group organization, machining, electrical setup, and design refinement. I learned how to approach practical build challenges, like tool limitations and time constraints, while still delivering useful, working results. Working on a real, functional vertical injection molding machine taught me the importance of iteration, team accountability, and balancing cost, complexity, and performance in engineering design.

6. Appendices

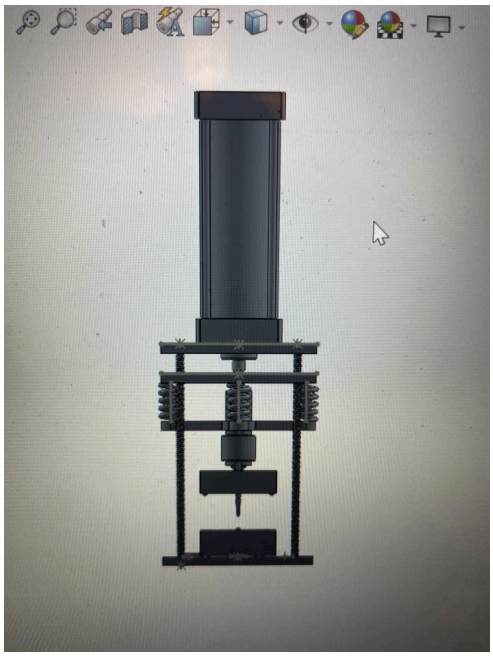


Figure 1: Version 0 of the Design



Figure 2: Version 1 Motor Threads

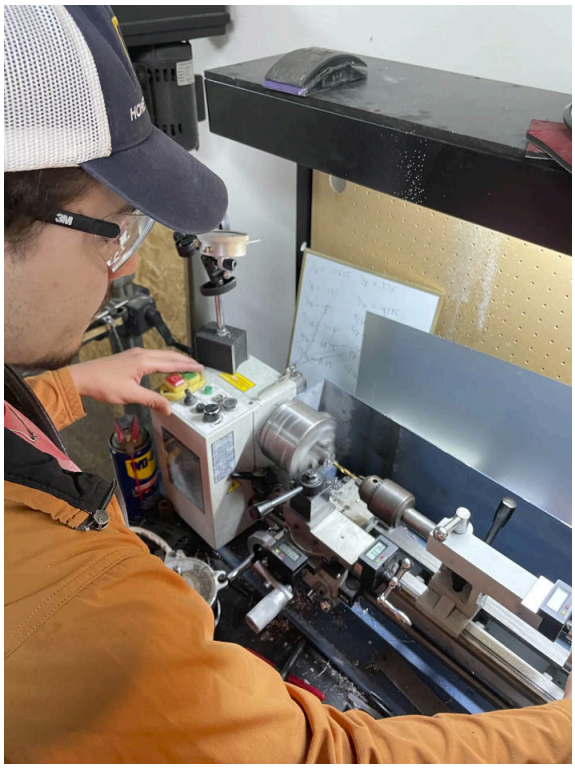


Figure 3: Machining the V1 Motor to Lead Screw Adapter with a Lathe



Figure 4: Testing the Tapping Tool



Figure 5: Tapping the V1 Motor to Lead Screw Adapter



Figure 6: Pneumatics Discussion



Figure 7: Pneumatic Testing

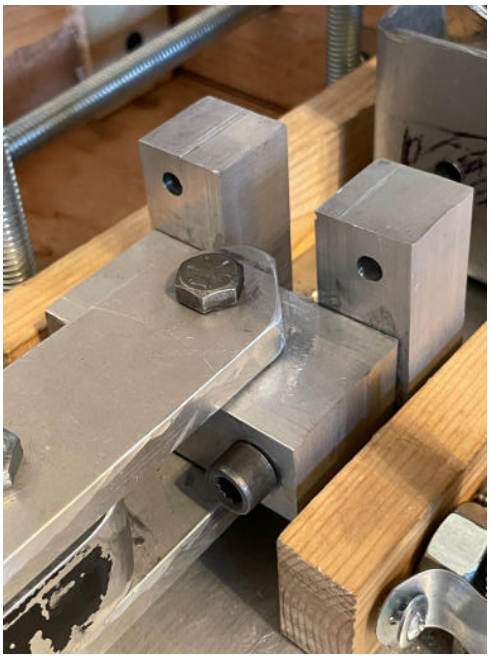


Figure 8: Clamp to Mold Adapter Rear View

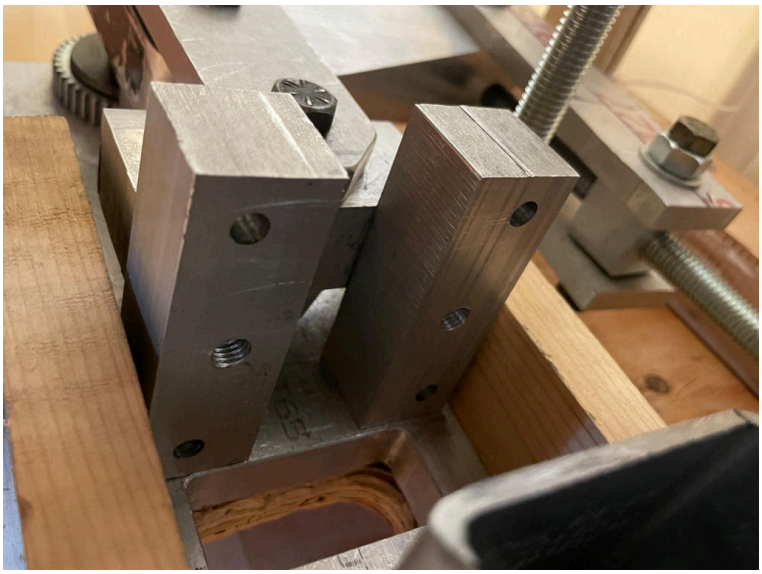


Figure 9: Clamp to Mold Adapter Front View



Figure 10: Step Drill Bits

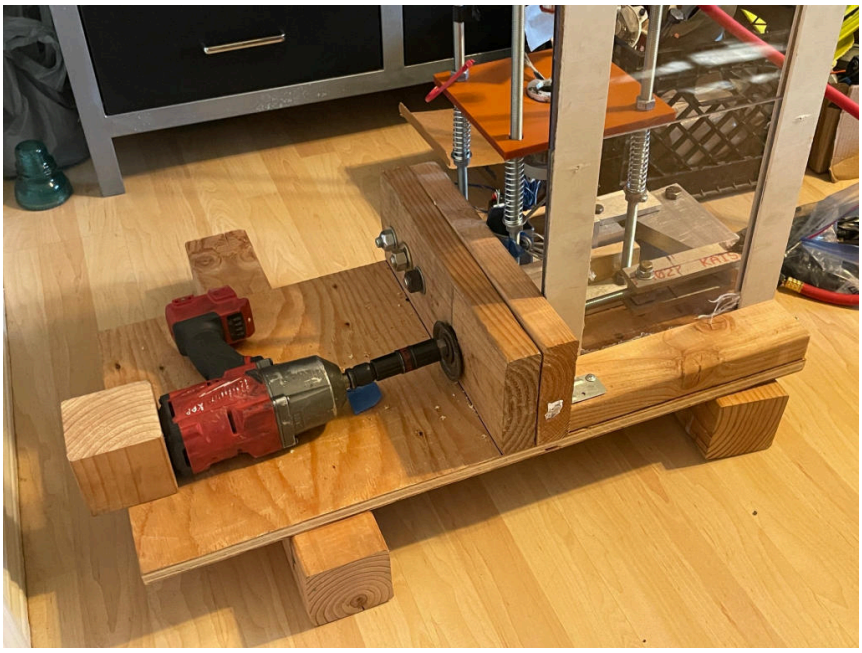


Figure 11: Wood Frame



Figure 12: Final Presentation

Nico Pence – Individual Contribution Report

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Abstract

Our group was tasked with making a fully automated vertical injection molding machine. I will cover the design and modeling for the plunger. Next is the design, analysis, and modeling for the clamp. There is some more minor additional help.

Introduction Short

First I will cover the general work I did in the first half of the project. Next is my design of the plunger. The plunger is how we referred to the part screwed on to the piston that touches and ejects the plastic through the barrel. After that is my design of the mechanical clamp. This is used to compress the mold to prevent the shot from escaping during injection. It is able to open and close with 3000 lb of injecting force. Lastly is primarily how I helped my group manufacture the clamp. The next section starts with a summary of my analysis, followed by the only nontrivial calculation used for the plunger. Then is a section with simplified calculations for the clamp. Ending my section is a conclusion on my work.

My Design and Contributions

First Test

For the first test my contributions were mostly in the design however most of this was collaborative so I'm only mostly sure these were my contributions. We all decided on the general parts. I was incharge of relaying the advice you gave in class like using the phenolic plate, that you will make the barrel for us, and any guidelines specified in class. During this design phase I mostly pointed out what potential designs wouldn't work and why so most of the design came from my groupmates, the video, or you and I only prevented potential issues. This resulted in a simple design that created a great shot once we ironed out the issues.

Plunger

The plunger that got machined was Plunger V4. This design attempted to be way too conservative towards the machinist. A diameter of 7/8th was chosen to prevent heated Aluminum Aluminum contact to the barrel and because I thought ample space would prevent it from being too hard to machine. I counted the threads and measured the piston to determine the threads were M20 x 1.5 and were 40mm long. While CADing I found the distance between the threads and 7/8in diameter was small. I was advised that would not be enough so I expanded the threaded section as shown below in Figure 1 and 2. The O ring was Parsa's Idea on how to seal the gap so earlier versions had a circular divot. Hudson explained that would be hard to machine so I went too far and made the divot a triangular cut. This is suboptimal as the O ring could slip easy, but the fit was tight enough that this was not the case. The other Version 4.5 fixed this by using a Square divot but this V4 was chosen by Hudson and I due to the ease of machining.

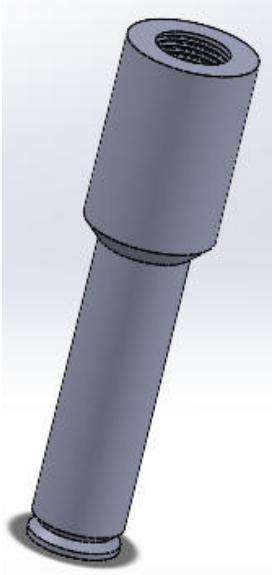


Figure 1:Plunger V4 Solidworks Model

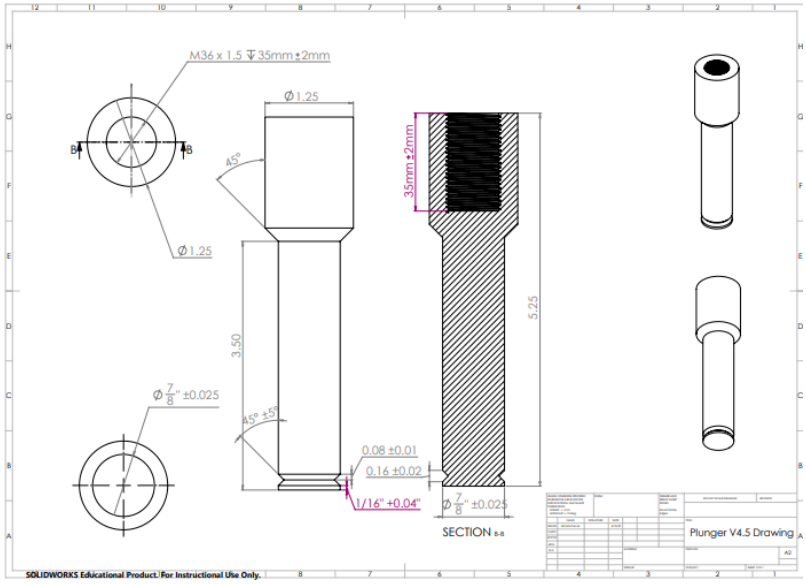


Figure 2: Plunger V4.5 Solidworks Drawing

Clamp

The Clamp was based on the concept shown in the motion study lab. That was a while back so the concept ended up slightly different.

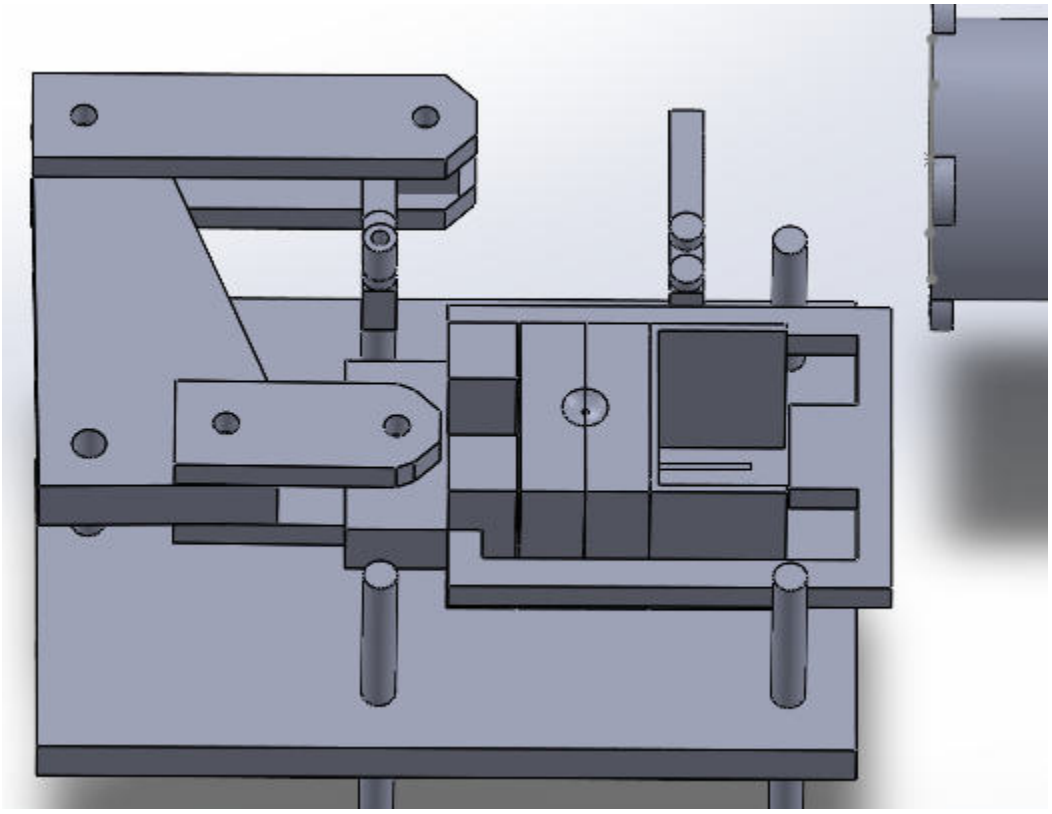


Figure 3: Clamp Solidworks Assembly Locked Position

In the locked position shown above in figure 3, the clamping force is theoretically entirely axial. The load travels from the mold to T shaped adapter to the shear pin through it (not shown) to the mold rods, then another shear pin, down the triangle to the last pin and through a larger shear pin to the plate. This when combined with the backstop that I did not design would create the 3000lb clamping force. In this position, the triangle, screw rods, and invisible lead screw only have the function of keeping the triangle from rotating and releasing the clamping force which is in an unstable equilibrium. As all the force goes directly to the triangle's large pin, the screw and rods should experience no force. Practically, There will be some but the resulting force should be minimal at lower angles. The appendix shows all calculations at a triangle angle of 5 degrees full 3000 lb clamping force.

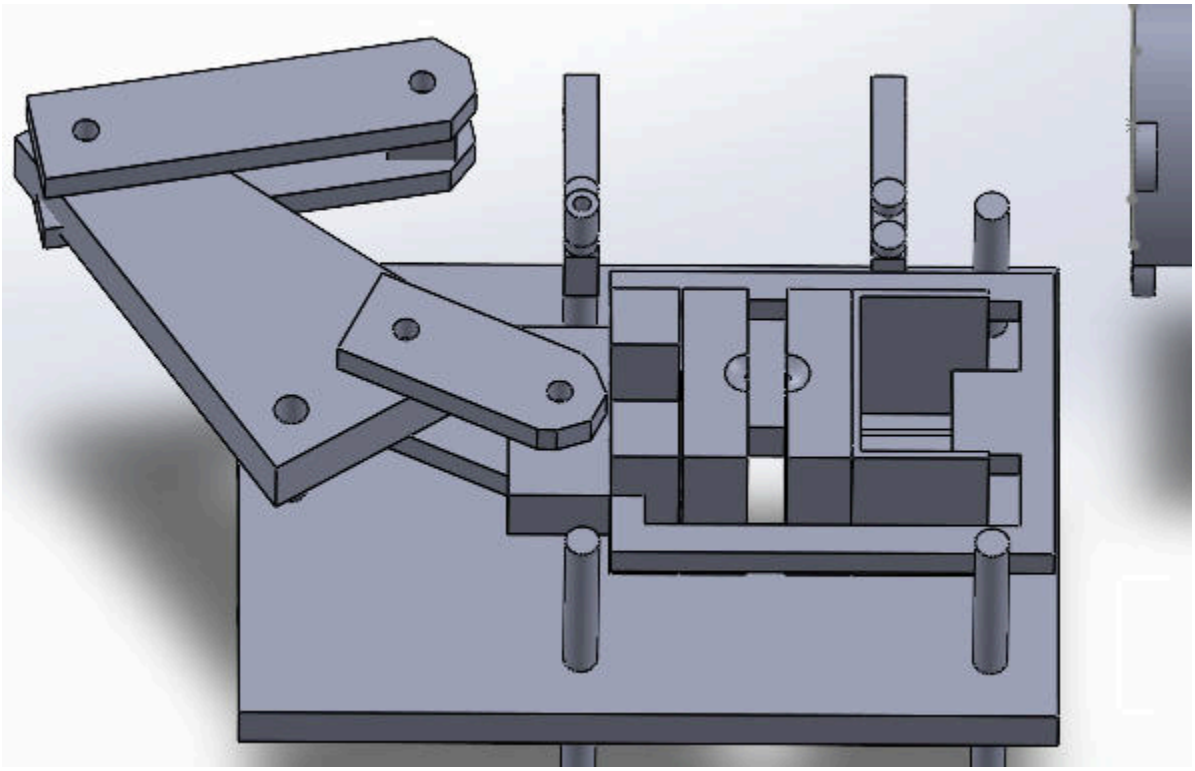


Figure 4: Clamp Solidworks Assembly Alternate Open Position

Above in figure 4: is the clamp in the open position. When the triangle is rotated (either direction works) it pulls the mold back. The purpose of the rod mold adapters is to allow the mold to slide without rotation. This would increase the force however, there is no clamping force while it is open. The assumption of minimal force while open and minimal force to the lead screw while closed is the reason issues like bending in the leadscrew can be mostly ignored.

Final Test

For the final test my contributions mostly pertain to overseeing the manufacturing of the clamps. I resized the parts to fit in the final design and have the proper static safety factors (mathcad made this easy once set up). I determined the sizes of the 2 plates my groupmates bought to machine the clamp out of partially shown in Appendix “Clamp Mathcad Plate Buying Analysis”. I altered the design to adjust for the ease of machining by my groupmates. This included removing unneeded curves or simplifying them, altering numbers or locations of holes in clamp parts and plates. A small amount of drawing cut and hole lines for some parts. The actual fabrication of the clamp was mostly my groupmates. Finding many of the correct bolts, washers, and nuts. Explaining the design and some sizing to attach the motor and leadscrew to the whole machine.

Analysis I Performed

Summary

The following contains a simple calculation for the plunger and after is the static analysis of the clamp. This includes Axial, buckling, and bolt shear/bearing

Plunger

When designing the triangular grove cross section I did area calculations

$O\ ring\ cross\ section\ dia\ d\ \simeq\ 0.137\ in$

$Space\ inbetween\ =\ \frac{D\ barrel - D\ plunger}{2} = \frac{1in - 7/8}{2} = \frac{1}{16} in\ =\ 0.0625 in$

As this was over half the O rings d, I assume putting half the O ring in the triangular would work

$O\ ring\ Area\ =\ \frac{\pi}{4} d^2 = \frac{\pi}{4}(0.137\ in)^2 = 0.0147 in^2$

Since the two half triangles can be folded into a square

$L\ try\ = \frac{L\ Square}{\sqrt{2}} = \frac{\sqrt{0.0147}}{\sqrt{2}} = 0.0858 in$ and I decided 0.08 in because it's easier to remove than add material. This is how I determined the dimension shown below in Figure 5.

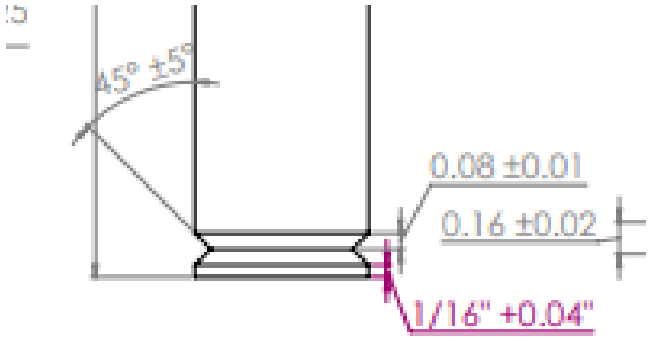


Figure 5: Zoom in of Plunger V4.5 drawing O ring divit

Clamp

As the clamp calculations become lengthy I will only show the most vulnerable section, between the triangle and the mold. The calculations will be in the perfectly locked position. Most of the calculations are shown in full at and angle of 5 degrees from closed is done in mathcad shown in the appendix.

All numbers are from pin to pin and a 1.5 times diameter (as specified by shigleys) is added to material to prevent it from failing.

$Triangle\ length\ =\ 2in,\ Rod\ length\ =\ 2in,\ Rods\ cross\ section\ is\ 0.5 in\ by$

The triangle thickness was chosen as 1in for easy procurement and the rods are half that 0.5in tall as it is in double shear. The material is Aluminum with a yield strength of 40 ksi. I decided to put

a 80% proof strength resulting in $40\ ksi\ *\ 0.8\ =\ 32\ ksi$ despite having a design factor goal of 2. When using 3/8in bolts annding 1.5 d more gets a width of

$W\ =\ d\ +\ 2\ *\ 1.5\ *\ d\ =\ d\ *\ d\ =\ 2\ *\ 3/8\ =\ 1.5\ in\ wide$

Cross sectional area = $0.75in\ *\ 0.5in\ =\ 0.375in^2$

As force is purely axial and split between them F is 1500 lb

$\sigma\ axial\ =\ F/A\ =\ 1500\ lb/0.375in^2\ =\ 4ksi$

$n\ axial\ =\ Available/Actual\ =\ 32\ ksi/4ksi\ =\ 8$

$Aluminum\ Elastic\ modulus\ =\ 10.4\ Mpsi\ +\ Rod\ E$

$I\ =\ bh^3/12\ =\ 1.5\ in\ *\ (0.5in)^3/12\ =\ 0.0165\ in^4$

For 2 pin supports $C\ =\ 1$

$Pcr\ =\ c\ *\ \pi^2\ *\ E\ *\ I/L^2\ =\ 1\ *\ \pi^2\ *\ 10.4E^6psi\ *\ 0.0165\ in^4/(5.5in)^2\ =\ 33.7\ kip$

$n\ buckling\ =\ Pcr/Load\ =\ 33.7kip/1.5kip\ =\ 22.5$

The factor of safety for buckling is recommended to be much higher since bucking failures are much more dangerous. 22.5 will be fine.

Bolts

$A\ =\ \frac{\pi}{4}d^2\ =\ \frac{\pi}{4}\left(\frac{3}{8}in\right)^2\ =\ 0.11\ i$, desired Factor of safety 2

$Bolt\ shear\ =\ (F\ *\ n)/(2\ *\ 0.6\ *\ A)\ =\ (3000lb\ *\ 2)/(2\ *\ 0.6\ *\ 0.11)\ =\ 45\ kis$

$Bolt\ bearing\ =\ (F\ *\ n)/(d\ *\ t)\ =\ (3000lb\ *\ 2)/(3/8in\ *\ 1in)\ =\ 16\ ksi$

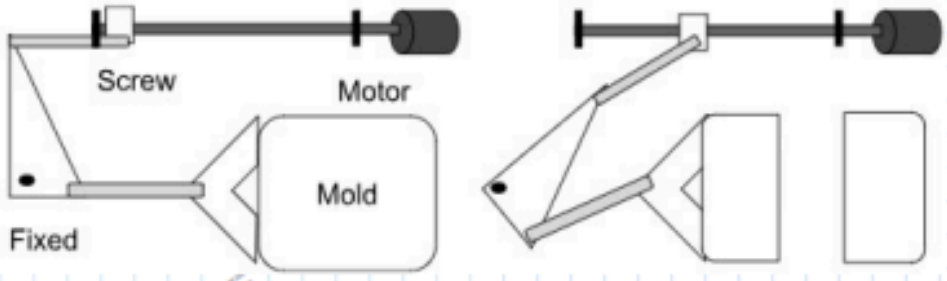
45 ksi requires a grade 4 or higher bolt by checking Shigley's graph shown in the mathcand in the appendix on page 12.

Conclusion

This project has taught me quite a bit about real mechanical engineering work. I saw how effective the engineering design process is when used and the issues that arise when not. I've learned some of what is easy or hard to machine and how this depends on the machinist and the tools they have access to. We saw the effect of lead times and how you handled them properly. We also did the opposite which limited options and increased time and money costs. I got much more practice with solidworks and learned from my group how to make engineering drawings. I learned the vast amount of analysis required to make even simple mechanisms work and all the decisions required to make it functional. Overall this project showed me some of what it's like to be a mechanical engineer and I am very grateful for it.

Appendices

Static Analysis



Position

Origin at pivot

$TryL := 2\ in \quad TryH := 3.5\ in + 2\ in \quad \theta t := -5\ °$ Triangle Dimentions

$TryMx(\theta) := TryL \cdot \cos(\theta)$ $TryMy(\theta) := TryL \cdot \sin(\theta)$ Triangle Point Equations

$TryMx(\theta t) = 1.992\ in$ $TryMy(\theta t) = -0.174\ in$

$TrySx(\theta) := -TryH \cdot \cos(\theta - 90\ °)$ $TrySy(\theta) := -TryH \cdot \sin(\theta - 90\ °)$ Triangle Side Equations

$TrySx(\theta t) = 0.479\ in$ $TrySy(\theta t) = 5.479\ in$

$RodML := 2.0\ in \quad RodMy := 0$ Rod/adapter joint Position

$RodMx(\theta) := TryMx(\theta) + \sqrt{RodML^2 - (TryMy(\theta))^2}$ $RodMx(\theta t) = 3.985\ in$

$AdaL := 2\ in \quad AdaMy := 0$ adapter/mold Position

$AdaMx(\theta) := RodMx(\theta) + AdaL \quad AdaMx(\theta t) = 5.985\ in$

$ML := 1.9\ in \quad MH := 3\ in \quad Mx(\theta) := AdaMx(\theta) + \frac{ML}{2}$ Mold Faces Positions

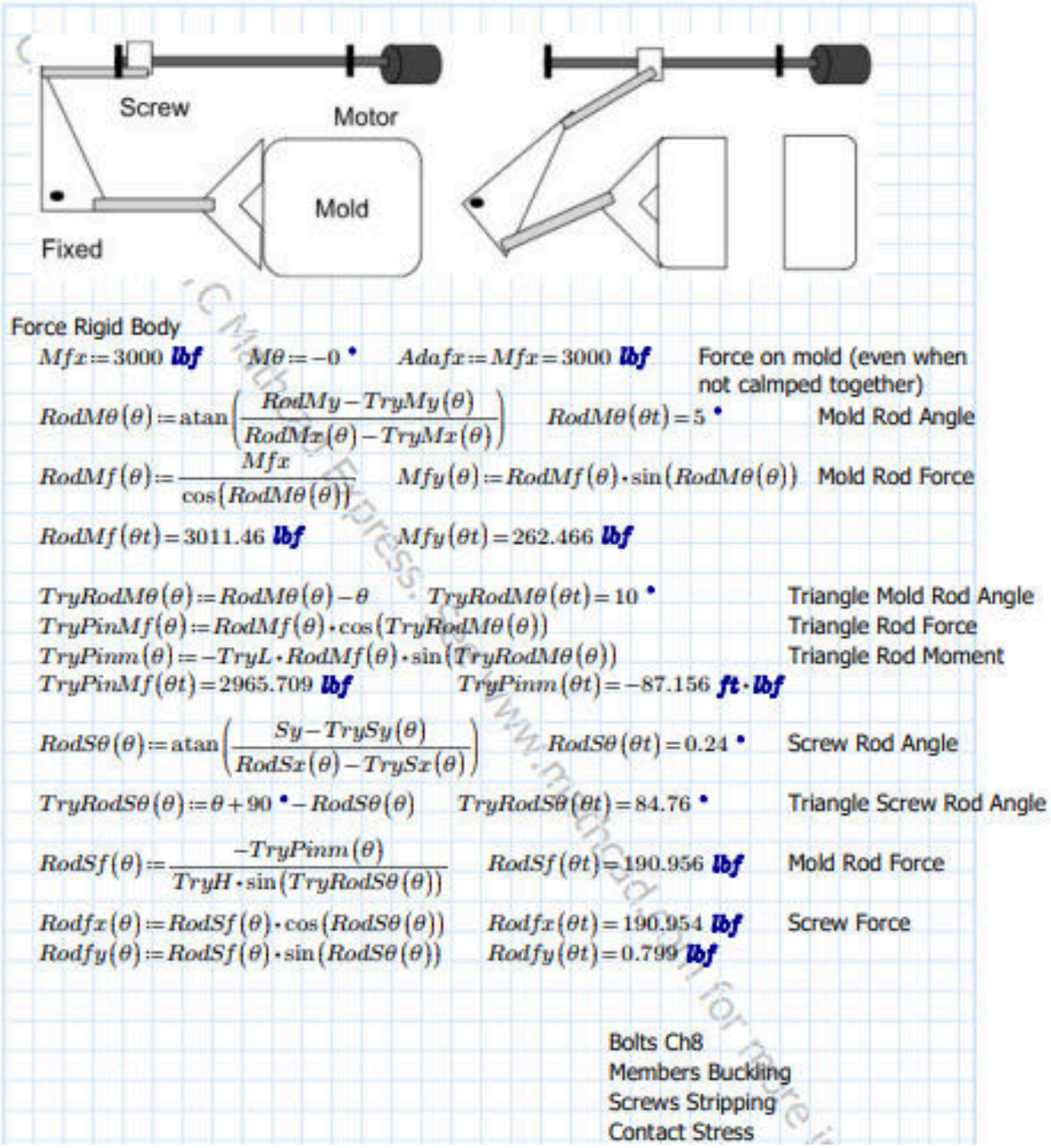
$MFx := Mx(0\ °) = 6.95\ in \quad Mx(\theta t) = 6.935\ in$

$RodSL := 5\ in \quad Sy := TryH = 5.5\ in$ Rod Screw Position

$RodSx(\theta) := TrySx(\theta) + \sqrt{RodSL^2 - (Sy - TrySy(\theta))^2}$ $RodSx(\theta t) = 5.479\ in$

$RodSx(0\ °) - RodSx(-40\ °) = -3.367\ in \quad Mx(0\ °) - Mx(-40\ °) = 0.936\ in$

Clamp Mathcad Position Analysis



Force Rigid Body

$Mfx := 3000\ lbf$ $M\theta := -0°$ $Adafx := Mfx = 3000\ lbf$ Force on mold (even when not calmped together)

$RodM\theta(\theta) := \operatorname{atan}\left(\frac{RodMy - TryMy(\theta)}{RodMx(\theta) - TryMx(\theta)}\right)$ $RodM\theta(\theta t) = 5°$ Mold Rod Angle

$RodMf(\theta) := \frac{Mfx}{\cos(RodM\theta(\theta))}$ $Mfy(\theta) := RodMf(\theta) \cdot \sin(RodM\theta(\theta))$ Mold Rod Force

$RodMf(\theta t) = 3011.46\ lbf$ $Mfy(\theta t) = 262.466\ lbf$

$TryRodM\theta(\theta) := RodM\theta(\theta) - \theta$ $TryRodM\theta(\theta t) = 10°$ Triangle Mold Rod Angle

$TryPinMf(\theta) := RodMf(\theta) \cdot \cos(TryRodM\theta(\theta))$ Triangle Rod Force

$TryPinm(\theta) := -TryL \cdot RodMf(\theta) \cdot \sin(TryRodM\theta(\theta))$ Triangle Rod Moment

$TryPinMf(\theta t) = 2965.709\ lbf$ $TryPinm(\theta t) = -87.156\ ft \cdot lbf$

$RodS\theta(\theta) := \operatorname{atan}\left(\frac{Sy - TrySy(\theta)}{RodSx(\theta) - TrySx(\theta)}\right)$ $RodS\theta(\theta t) = 0.24°$ Screw Rod Angle

$TryRodS\theta(\theta) := \theta + 90° - RodS\theta(\theta)$ $TryRodS\theta(\theta t) = 84.76°$ Triangle Screw Rod Angle

$RodSf(\theta) := \frac{-TryPinm(\theta)}{TryH \cdot \sin(TryRodS\theta(\theta))}$ $RodSf(\theta t) = 190.956\ lbf$ Mold Rod Force

$Rodfx(\theta) := RodSf(\theta) \cdot \cos(RodS\theta(\theta))$ $Rodfx(\theta t) = 190.954\ lbf$ Screw Force

$Rodfy(\theta) := RodSf(\theta) \cdot \sin(RodS\theta(\theta))$ $Rodfy(\theta t) = 0.799\ lbf$

Bolts Ch8
Members Buckling
Screws Stripping
Contact Stress

Clamp Mathcad Force Analysis

Rods

Axial

$TryTh := 1\ in$ $RodMTh := 0.5\ TryTh$ $RodSTh := 0.5\ TryTh$ Member thickness

$RodMWi := TryTh = 1\ in$ $RodSWi := RodMWi$ Member Width

$RodMXa := 2 \cdot RodMTh \cdot RodMWi = 1\ in^2$ $RodSXa := 2 \cdot RodSTh \cdot RodSWi = 1\ in^2$

$RodM\sigma a(\theta) := \frac{RodMf(\theta)}{RodMXa}$ $RodS\sigma a(\theta) := \frac{RodSf(\theta)}{RodSXa}$ $RodM\sigma a(\theta t) = 3.011\ ksi$

$RodS\sigma a(\theta t) = 0.191\ ksi$

$TryYs := 40\ ksi$ $RodMYs := TryYs$ $RodSYs := TryYs$ Yeild strengths

$Proof := 80\%$ $RodMPs := Proof \cdot RodMYs = 32\ ksi$ $RodSPs := Proof \cdot RodSYs = 32\ ksi$

$RodMna(\theta) := \frac{RodMPs}{RodM\sigma a(\theta)}$ $RodMna(\theta t) = 10.626$ Factor of safety against Axial

$RodSna(\theta) := \frac{RodSPs}{RodS\sigma a(\theta)}$ $RodSna(\theta t) = 167.578$

$Rodnmin := min(RodMna(\theta t), RodSna(\theta t)) = 10.626$

Contact

$Try\nu := 0.333$ $RodM\nu := Try\nu$ $RodS\nu := RodM\nu$ $BoltM\nu := 0.27$ $BoltS\nu := BoltM\nu$ Constants

$TryE := 10.4 \cdot 10^6\ psi$ $RodME := TryE$ $RodSE := TryE$ $BoltME := 29 \cdot 10^6\ psi$ $BoltSE := BoltME$

$BoltNomd := \frac{3}{8}\ in$ $Rodd := \frac{25}{64}\ in = 0.391\ in$ $Boltd := BoltNomd - 0\ in = 0.375\ in$

$b(F, L, \nu_1, \nu_2, E_1, E_2, d_1, d_2) := \sqrt{\frac{2 \cdot F}{\pi \cdot L} \cdot \frac{(1-\nu_1^2) \cdot E_1^{-1} + (1-\nu_2^2) \cdot E_2^{-1}}{d_1^{-1} + d_2^{-1}}}$ Contact Stress

$RodMb := b(0.5 \cdot RodMf(\theta t), RodMTh, RodM\nu, BoltM\nu, RodME, BoltME, -Rodd, Boltd) = 0.046\ in$

$RodSb := b(0.5 \cdot RodSf(\theta t), RodSTh, RodS\nu, BoltS\nu, RodSE, BoltSE, -Rodd, Boltd) = 0.012\ in$

$C\tau(F, b, L) := 0.3\ \frac{2 \cdot F}{\pi \cdot b \cdot L}$ $RodMC\tau := C\tau(0.5 \cdot RodMf(\theta t), RodMb, RodMTh) = 12.518\ ksi$

$RodSC\tau := C\tau(0.5 \cdot RodSf(\theta t), RodSb, RodSTh) = 3.152\ ksi$

$RodMnc(\theta) := \frac{RodMPs}{RodMC\tau}$ $RodMnc(\theta t) = 2.556$

Factor of safety against Contact

$RodSnc(\theta) := \frac{RodSPs}{RodSC\tau}$ $RodSnc(\theta t) = 10.152$

$Rodnmin := min(Rodnmin, RodMnc(\theta t), RodSnc(\theta t)) = 2.556$

Buckling

$C := 1$ $I(b, h) := min\left(\frac{b \cdot h^3}{12}, \frac{h \cdot b^3}{12}\right)$ $RodMI := I(RodMTh, RodMWi) = 0.01\ in^4$

$RodSI := I(RodSTh, RodSWi) = 0.01\ in^4$

$RodMPcr := 1 \cdot \pi^2 \cdot RodME \cdot RodMI \cdot RodML^{-2} = 267.302\ kip$

$RodSPcr := 1 \cdot \pi^2 \cdot RodSE \cdot RodSI \cdot RodSL^{-2} = 42.768\ kip$

$RodMnb := \frac{RodMPcr}{RodMf(\theta t)} = 88.762$ $RodSnb := \frac{RodSPcr}{RodSf(\theta t)} = 223.969$ Factor of safety against Buckling

$Rodnmin := min(Rodnmin, RodMnb, RodSnb) = 2.556$

Created with PTC Mathcad Express. See www.mathcad.com for more informa

Clamp Mathcad Rods Factor of safety Analysis

Bolts

Keep them 1.5 diameters from the edge

$BoltMYs := TryYs$ $BoltSYs := TryYs$ Yeild strengths

$Proof := 80\%$ $BoltMPs := Proof \cdot BoltMYs = 32$ ksi $BoltSPs := Proof \cdot BoltSYs = 32$ ksi

$$BoltMr\sigma := \frac{RodMf(\theta t)}{2 \cdot min(TryTh, 2 \cdot RodMTh) \cdot Boltd} = 4.015 \text{ ksi} \qquad BoltMn\sigma := \frac{BoltMPs}{BoltMr\sigma} = 7.97$$

$$BoltSr\sigma := \frac{RodSf(\theta t)}{2 \cdot min(TryTh, 2 \cdot RodMTh) \cdot Boltd} = 0.255 \text{ ksi} \qquad BoltSn\sigma := \frac{BoltSPs}{BoltSr\sigma} = 125.683$$

$Boltnmin := min(BoltMn\sigma, BoltSn\sigma) = 7.97$

$$BoltA := \frac{\pi}{4} \cdot BoltNomd^2 = 0.11 \text{ in}^2 \qquad Boltn := 2 \qquad SC = \frac{2 \cdot 0.6 \cdot BoltUt \cdot A}{n}$$

Factor of safety against Bolt Shear

$$BoltUts := \frac{RodMf(\theta t) \cdot Boltn}{2 \cdot 0.6 \cdot BoltA} = 45.444 \text{ ksi}$$

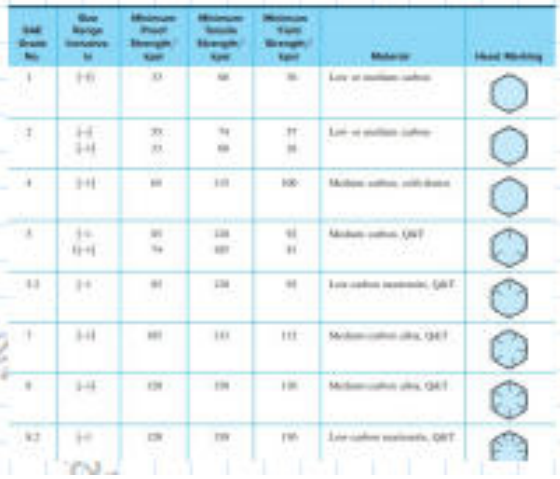
$$BoltUtb := \frac{RodMf(\theta t) \cdot Boltn}{BoltNomd \cdot TryTh} = 16.061 \text{ ksi}$$

$\max(BoltUts, BoltUtb) = 45.444$ ksi

Grade 2+ but might as well go higher

$$BoltA2 := \frac{\pi}{4} \cdot (0.5 \text{ in})^2 = 0.196 \text{ in}^2$$

$$BoltUts := \frac{RodMf(\theta t) \cdot Boltn}{0.6 \cdot BoltA2} = 51.124 \text{ ksi}$$



Clamp Mathcad Bolts Factor of safety Analysis

Triangle

$TryMb := b(RodMf(\theta t), TryTh, Try\nu, BoltM\nu, TryE, BoltME, -Rodd, Boltd) = 0.046$ in

$TrySb := b(RodSf(\theta t), TryTh, Try\nu, BoltS\nu, TryE, BoltSE, -Rodd, Boltd) = 0.012$ in

$TryPs := Proof \cdot TryYs = 32$ ksi $TryMC\tau := C\tau(RodMf(\theta t), TryMb, TryTh) = 12.518$ ksi

$TrySC\tau := C\tau(RodSf(\theta t), TrySb, TryTh) = 3.152$ ksi

$$TryMnc := \frac{TryPs}{TryMC\tau} = 2.556$$

Factor of safety against Contact

$$TrySnc := \frac{TryPs}{TrySC\tau} = 10.152$$

$Trynmin := min(TryMnc, TrySnc) = 2.556$

Clamp Mathcad Triangle Factor of safety Analysis

Plate Sizing

$Ed := BoltNomd \cdot \left(\frac{1}{2} + 1.5\right) = 0.75\ in$

Try	$TryL + 2 \cdot Ed = 3.5\ in$	$TryH + 2 \cdot Ed = 7\ in$
2xRodM	$RodML + 2 \cdot Ed = 3.5\ in$	$2 \cdot Ed = 1.5\ in$
2xRodS	$RodSL + 2 \cdot Ed = 6.5\ in$	$2 \cdot Ed = 1.5\ in$
Adapter	3x 1 *in*	3 *in*
Screw Adapter	?	1 *in*

$3.5 + 1.5 \cdot 2 + 1.5 + 1 + 1 = 10$ 7x11 8x12

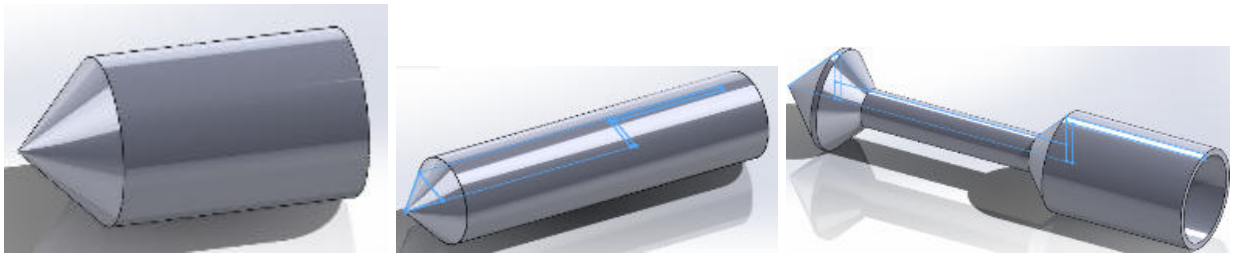
8+in by 12in by 1 in aluminum plate

$8\ in - Ed - TryL - RodML - AdaL - 0.5 \cdot ML = 0.3\ in$

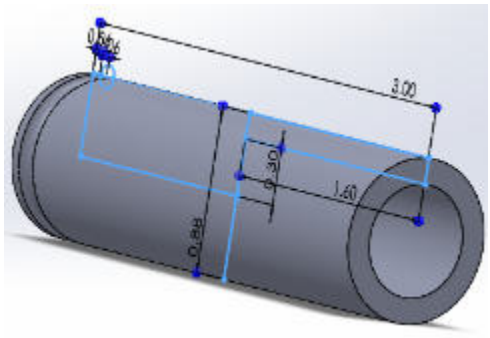
$8\ in - Ed - MFx = 0.3\ in$

Created with PTC Mathcad Express. See

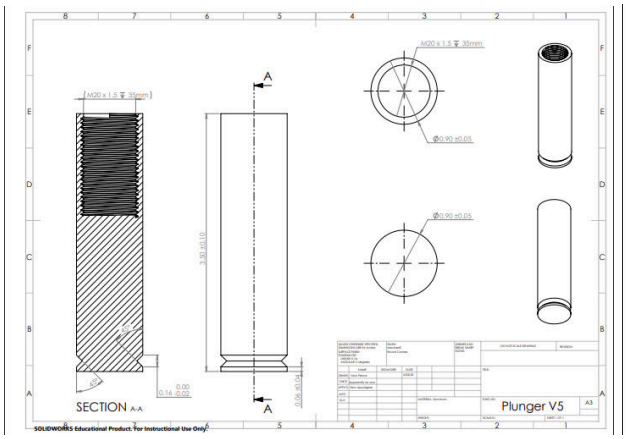
Clamp Mathcad Plate Buying Analysis

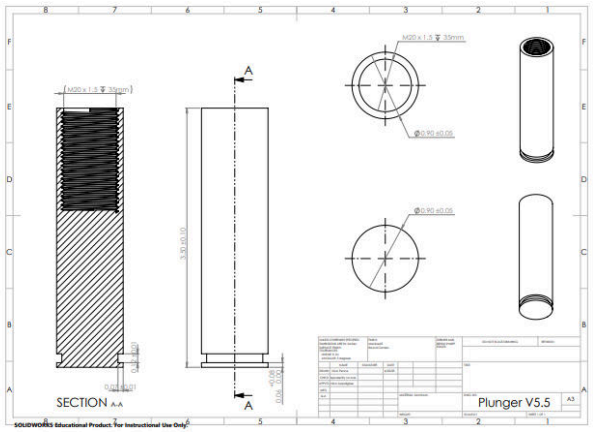


Version 1 Plungers

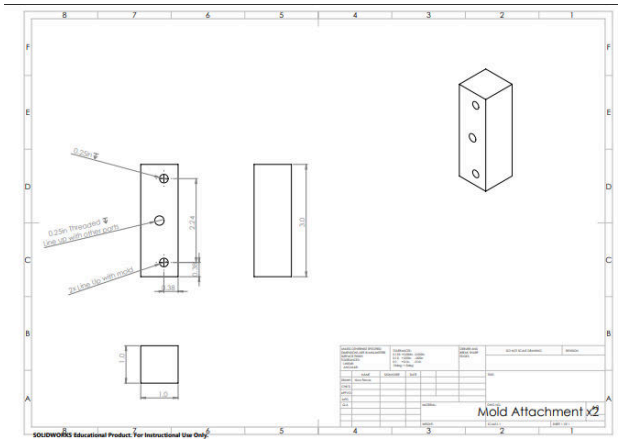


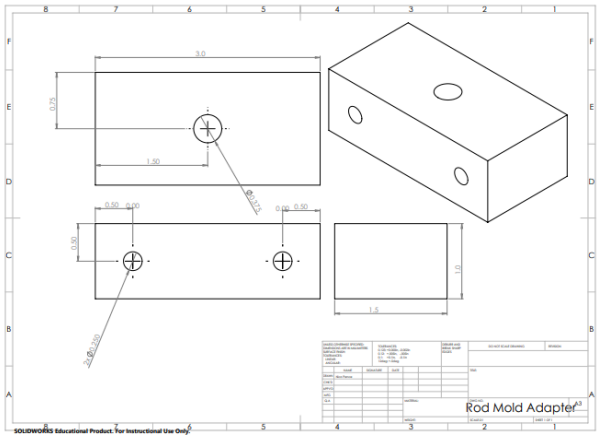
Version 2



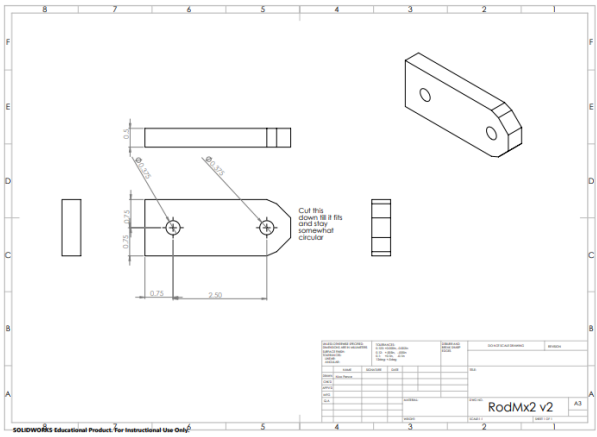


Version 5

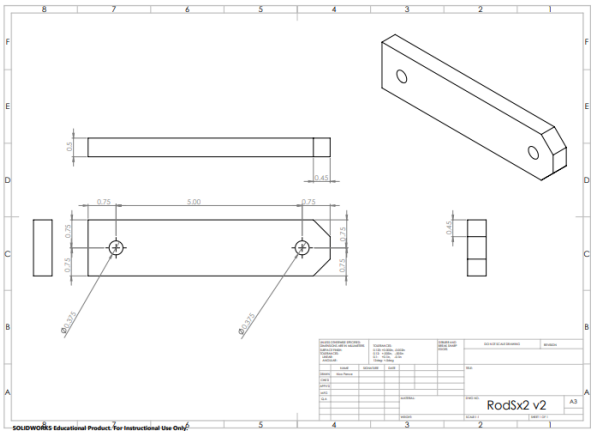




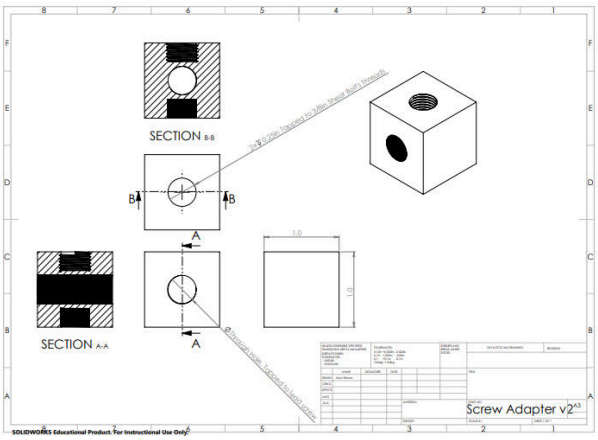
Clamp Mold Adapter



Mold Rods

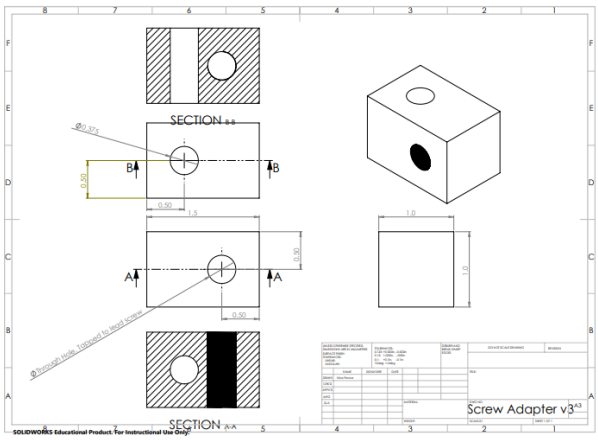


Screw Rods

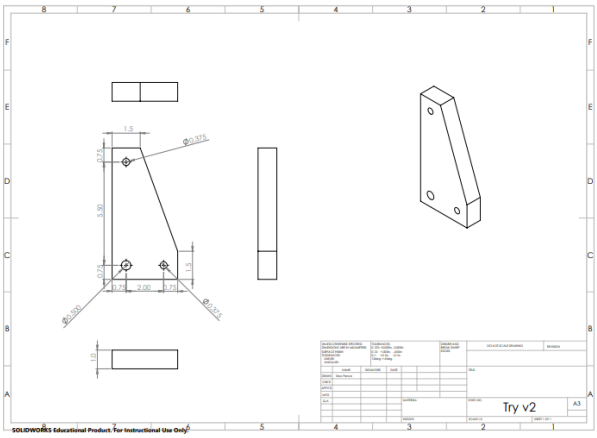


One we used

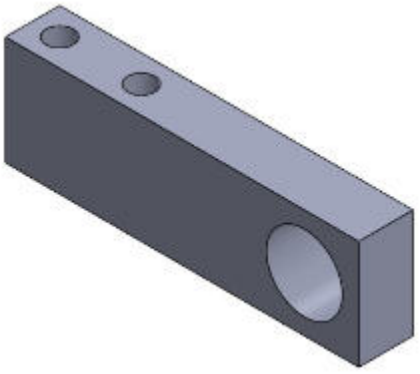
Lead Screw Adapters



One we didn't use



Clamp Triangle



Potential Lead Screw Holder

James Willis – Individual Contribution Report

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Page 2-3. Introduction, Design Contribution

Page 4. Analysis, Conclusion

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Abstract

For this project, my primary contribution was machining the clamp system that enables the injection process. I manufactured nearly all components of the clamp, including precise drilling, milling, cutting, and tapping—especially for the adapter and mating parts. The tapping and alignment process was time-intensive and critical to proper function. I also assembled our Gantt chart (Figure 7), organized both the PDR and CDR presentations, and provided input on design decisions such as the hopper system and initial sizing of the vertical support rods. During build days, I took initiative in troubleshooting misalignments and fabrication challenges, often machining solutions on the spot to keep us on track. My contributions were essential to both the structural and functional performance of the machine.

Introduction

Our team's goal was to design and build a functional plastic injection molding machine using a mix of custom and off-the-shelf parts. My role centered on the mechanical side of the project, especially the fabrication of the clamp system (Figures 1–5). I also contributed to project organization, documentation, schedule keeping, CAD assembly (Figure 6), and successful execution of key milestones within the project while providing key input into our team designs and troubleshooting throughout the execution of the project.

Design & Contribution

Project Planning and Documentation:

I created the project Gantt chart (Figure 7), which outlined our milestones and helped the team stay on track and added to my role of keeping the team organized and on schedule, ensuring that all required designs had been completed and approved, parts were ordered, assembly was complete and tests were successful. I also assembled our PDR and CDR presentations, ensuring we had clear visuals and explanations for our evolving design.

Design Input and Group Collaboration:

I provided input during design discussions—particularly regarding the hopper feeding system, operation of the moving, spring-supported middle plate, and support rod configuration. I participated in early force estimation and sizing of the ½-inch diameter, 2-foot-long threaded rods used to support the machine and sustain the loads from mounted components. I contributed to the design of the hopper feeding system, ensuring that the rotating shot delivery device dispensed an adequate volume of ABS plastic into the mold. I also worked on integrating a motor to rotate the delivery system and contributed to the CAD assembly in SolidWorks (Figure 6), making sure that all independently drawn parts fit and functioned together as intended.

Clamp System Machining:

My main contribution was fabricating the entire clamp system. This included:

- Milling and cutting the aluminum members (excluding the rough initial sizing done by others)
- Drilling and tapping holes for the lead screw adapter (Figure 3), member connection joints (Figure 1), guide supports, and bottom plate
- Ensuring squareness and proper alignment to avoid binding during motion, using precise milling (Figure 4)
- Adjusting dimensions where needed for proper fit and performance

After the clamp members were cut to rough size—two 4" × 1.5" members, two 6.5" × 1.5" members, one triangular rotating piece, and one 1" × 1" cube—I took full responsibility for finishing their machining. Each required two ⅜" holes located ¾" from the edge for proper assembly. The triangular piece required a ½" hole for mounting to the base plate, which also had to be drilled. Milling was required to finalize part dimensions (Figure 4). I also machined mounting holes for the backstop and injector pin and tapped all required threads to fasten the parts. The full clamp assembly after completion of my fabrication work is shown in Figure 5.

Assembly and Troubleshooting:

I led troubleshooting during final assembly. When we encountered misalignments or hardware issues, I machined on-the-fly fixes to keep progress moving. I also supported teammates with their machining tasks and offered guidance during critical build days.

Analysis

Most of my analysis work was hands-on: checking alignment of tapped holes, verifying smooth motion of the clamp under torque, double-checking design calculations, and confirming that fabricated parts met design intent. I also assisted in calculating the volume of ABS delivered through the rotating hopper mechanism to ensure sufficient material was injected. Pneumatics system testing and clamp behavior during injection were observed and documented during trials (Figure 8).

Conclusion

I contributed heavily to the machining and assembly of our injection molding machine, particularly the clamp system, which is a core part of its mechanical function. Most other assemblies and components were off-the-shelf and required minimal fabrication, allowing us to keep costs around $50 per person. In contrast, the clamp system required extensive manual machining and was fabricated almost entirely by me from raw aluminum stock. I also played a supporting role in planning, documentation, CAD, and group design discussions. This project allowed me to strengthen both my fabrication skills and my ability to solve practical engineering challenges under tight time and budget constraints.

Appendices

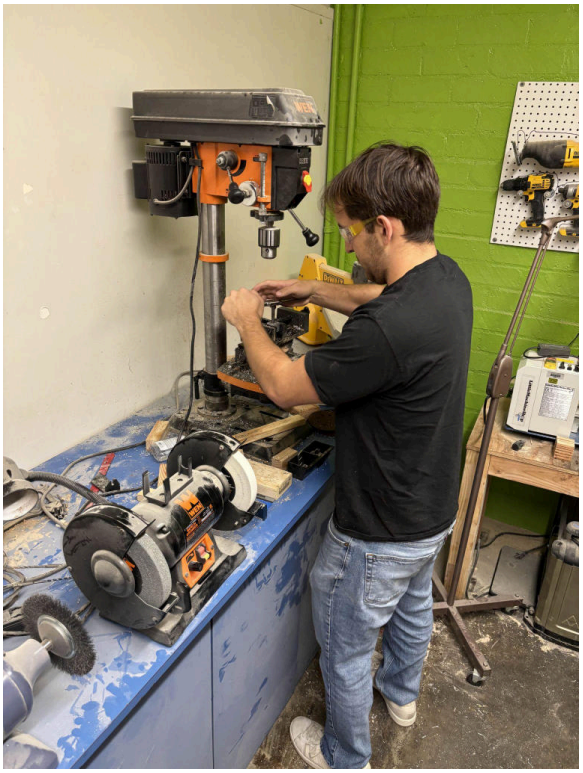


Figure 1. Drilling of connection holes in clamp members

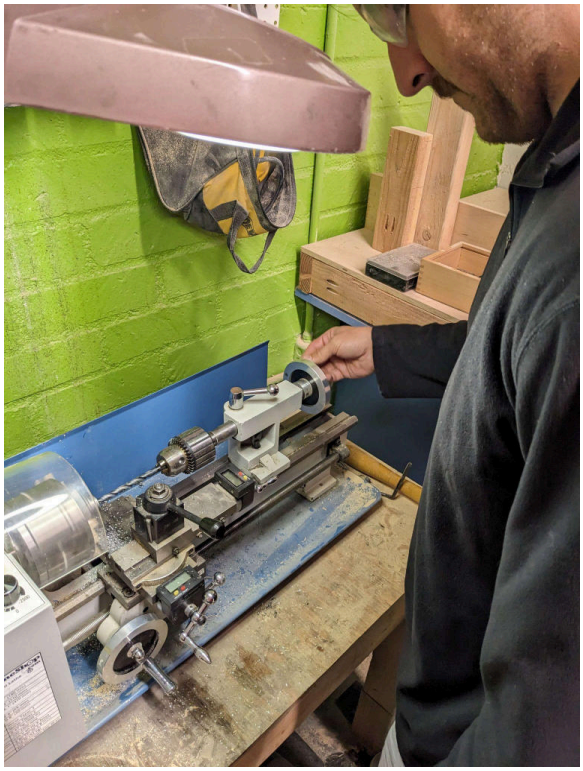


Figure 2. Using lathe to make lead screw adapter



Figure 3. Tapping lead screw connection point



Figure 4. Milling clamp swivel member to ensure 20 degree angle

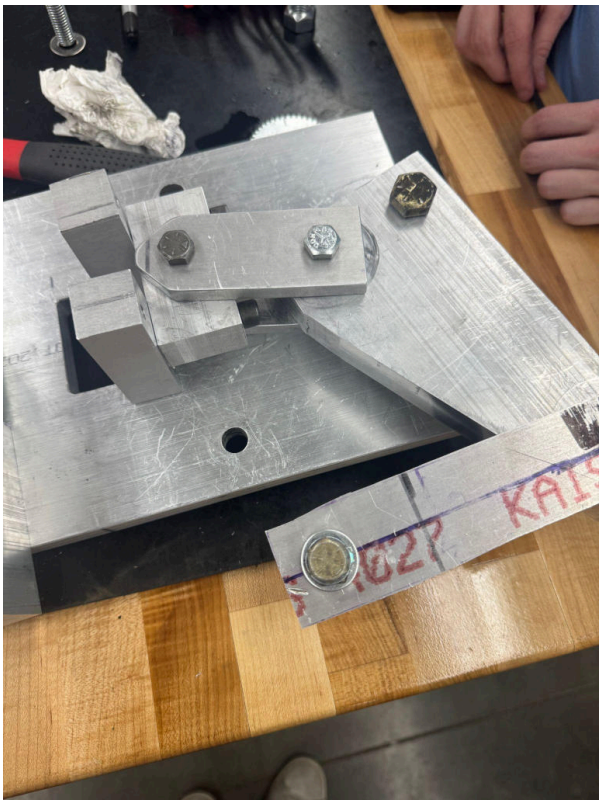


Figure 5. Full Assembled clamp after i had manufactured all members

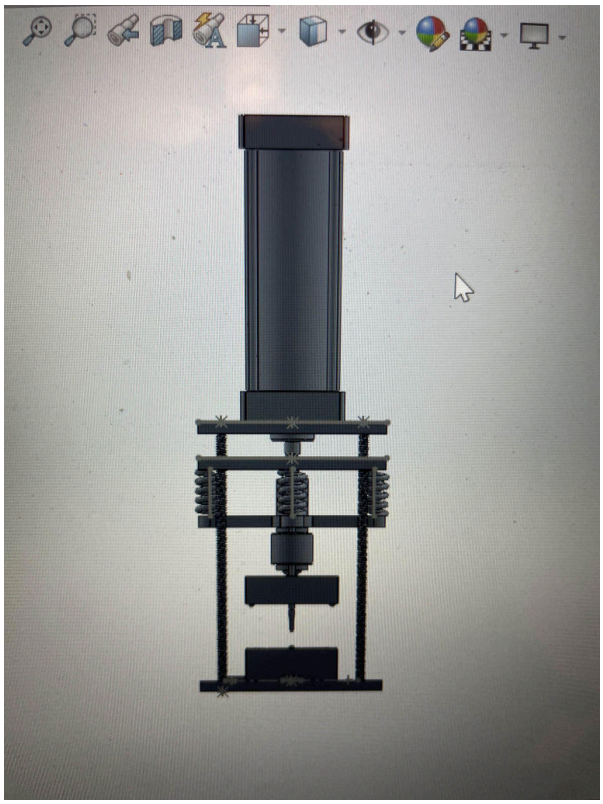


Figure 6 Version 0 CAD assemble

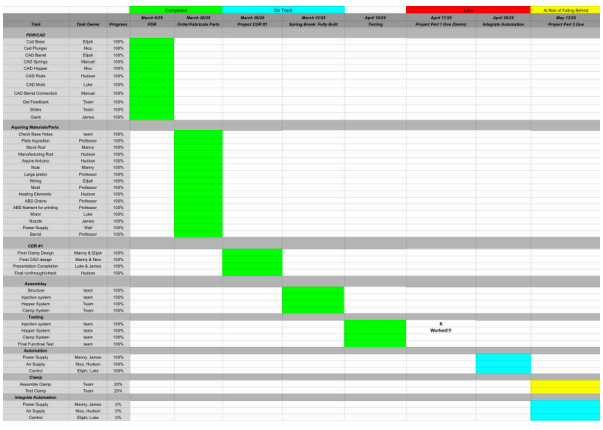


Figure 7 Gantt Chart



Figure 8. Pneumatics testing/demonstration #1 testing

Manuel Francisco Gonzalez
Group 2, Section 4 – ME 3250 Final Project Report

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   3.4 Build Process.....4

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Abstract

My contribution to Group 2, Section 4's Machine Design Final Project, consisted of the following in no particular order:

- Initial Conceptual Design Brainstorming
- Sketches and ideas for the layout of our initial design
- Financial Contributions
- Outsourcing:
    - Aluminum material for machining
    - Assembly pieces (nuts, bolts, washers etc.)
    - Injection nozzle tip
    - Threaded Rod for our vertical design
    - Phenolic Plastic Board
- Build time and tools for building off-campus
- Injection force calculations
- Scrapped clamp ideas

Additionally, I actively participated in the build phase at off-campus facilities, conducted simple injection force calculations, and created SolidWorks assemblies for our preliminary and critical design reviews.

Introduction

The project objective was the comprehensive design, fabrication, and automation of a vertical plastic mold injector. This section of the report details my roles,as I performed them. My responsibilities included mechanical design and analysis, detailed CAD modeling, cost-effective material sourcing, and hands-on assembly and fabrication.

Your Design/Contribution

3.1 Final Model Designs

The design process began with initial sketches and conceptual brainstorming. These early stages aimed to visualize our project's requirements. The initial finalized concept was a three-section vertical mold injector, incorporating a piston cylinder and feeding system on top, a heating chamber and nozzle in the middle, and initially, a horizontally oriented mold at the bottom as shown in Figure 1. My contributions included comprehensive SolidWorks modeling of these components, initially presented at our Preliminary Design Review (PDR).

However, post-PDR, we identified critical limitations in the horizontal mold configuration. This realization prompted design revisions, leading to a vertically oriented mold to better manage the forces generated during injection. I undertook revisions to our CAD models, transitioning our designs to accommodate the vertical mold orientation.

The bulk of the project laid in the design of the clamping system which underwent many iterations before settling on one final design. My teammate Elijah and I were the ones who brainstormed most of the clamp ideas early on, but it was Nico's final idea that we all agreed upon.

The clamp design for our CDR presentation is shown in Figure 3 as a locking cam design. A driver gear, driven by a 40v motor, spins two gears attached to two cams pushing the mold in to place and locking it in. After criteria from our CDR presentation, the switch to a lead screw design was made.

3.2 Thermocouple and PID Controller Wiring

The PID controller was the most important part for the heating system of the final design. We ultimately settled on a cheap amazon PID controller that came with its own schematic, relay,

and thermocouple. We had also ordered two heating bands to connect to the barrel for melting the plastic pellets.

I had very little experience in following schematics and wiring/soldering, but with the help of teammate Elijah we were able to follow the schematic and successfully wire the heating bands we had in series. This decision was changed during our first demonstration and the bands were wired in parallel. This was the only change made meaning I had a successful first wire.

3.3 Outsourced Components

Recognizing the need for managing low budgets, I managed the procurement of needed materials and components for relatively cheap. I sourced aluminum materials, injection nozzle tips, phenolic plastic boards, and a variety of standard mechanical fasteners (nuts, bolts, washers, and threaded rods). Each selection was guided by the dual criteria of quality and cost-effectiveness, ensuring that the project remained within budget constraints while maximizing mechanical performance.

3.4 Build Process

The assembly and fabrication phase was executed primarily at off-campus facilities, utilizing an array of machining and fabrication tools. My responsibilities during the build included machining specific components from raw aluminum stock, precise assembly of the injector components, and resolving technical challenges encountered throughout the build. My hands-on contributions were limited to putting together the pieces acquired and in transitioning our theoretical designs into a functional prototype.

Analysis

4.1 Injection Force Analysis

The injection force analysis was critical for ensuring the structural integrity and performance of the injector. This analysis involved calculating the necessary injection force based on the required injection pressure and the cross-sectional area of the piston cylinder. The fundamental equation guiding this calculation was projected to be over 3000 lbf based on the inner diameter of the cylinder

4.2 Volume Calculations

Accurate volume calculations were performed to determine the necessary dimensions of the injection chamber and mold cavity. This ensured adequate volume for the plastic material while optimizing the size and efficiency of the injector. Calculations involved straightforward volumetric equations based on the geometry of cylindrical and rectangular components, ensuring precision in component dimensions and material usage. Detailed volume calculations and their implications on the injector design can be found in Appendix B.

Conclusion

In summary, my comprehensive contributions significantly influenced the successful outcome of our vertical plastic mold injector project. My role encompassed extensive design and modeling efforts, critical analysis, careful procurement of materials, and active participation in hands-on assembly and fabrication tasks. The iterative process of designing, analyzing, and building provided valuable insights and practical knowledge in machine design principles. Ultimately, these contributions ensured the team met project goals, and I learned a lot about working with a team, and the process it takes to design and automate a system.

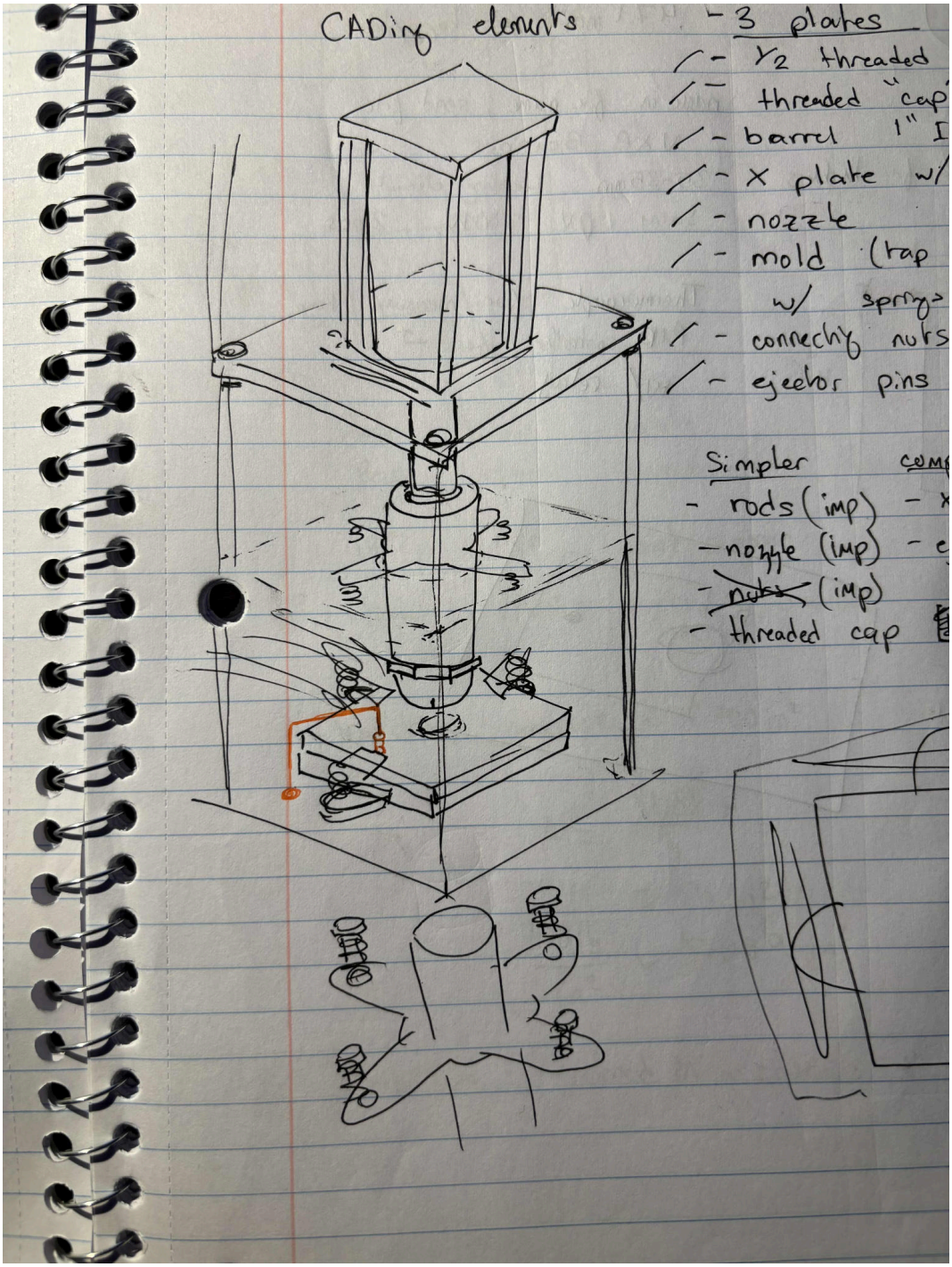


Figure 1: Early sketch of our initial design which ended up in our CDR presentation

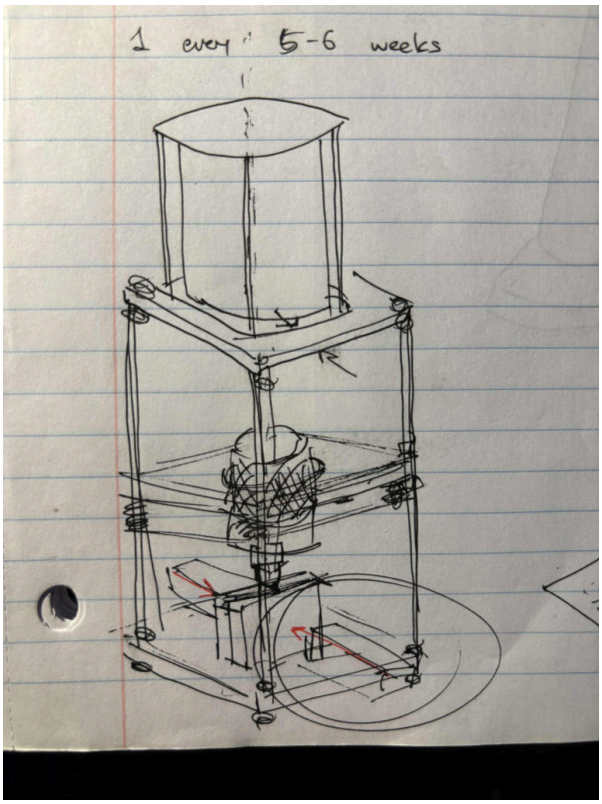


Figure 2: Secondary Design without the clamp design.

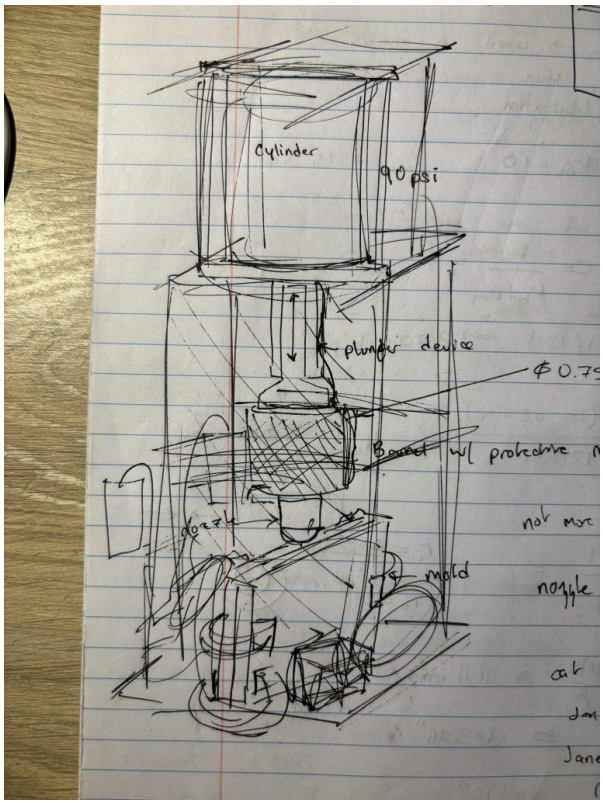


Figure 3: Secondary Design without the clamp design.

Hudson Brann – Individual Contribution Report

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1 Abstract

I led hands-on fabrication and rapid-response problem-solving for our desktop plastic-injection mold machine. Key deliverables include a wood-framed polycarbonate safety shield, a funnel produced within the hour after a 3-D-print failure, procurement and installation of the heater bands with a matching thermistor control, and directing the build of the manual pneumatic system. Also attempted to build the automatic pneumatic system, but was stopped short when our solenoid failed. I also machined: the plates, threaded rods, phenolic plate, and the steel coupling. I was our team's guiding light when things would go south. **(Fig.6.)** I fixed the electronics when they were wired incorrectly on demo day. I taught my teammates how to machine and good practices for machining.

2 Introduction

The goal of our project was to build a functional vertical injection molding machine that could operate basic thermoplastic molding processes at the cheapest cost to ourselves. Our work used CAD design, machining, electrical engineering, and safety. Our team made an effective manual injection machine, and had the potential for a great automatic injection all at a very low cost. **(Fig. 8)**

3 Design & Fabrication Contributions

3.1 Polycarbonate Shield (Fig. 1)

Materials: 6 mm clear PC sheet, 18 mm birch plywood frame, #10-24 × 1-in screws.

Design-for-assembly: Flat strips of wood were cut out on a band sa,w and I sandwiched the two plates between them, which allowed me to have removable sheets.

3.2 Plastic Bead Funnel (Fig. 2, 11, 12, 13)

Situation: PLA funnel print failed mid-run

Action: Used the materials we had on hand (ducttape, the failed print, cardboard) and made an attempt to deliver the beads into the funnel. I had coded and built the electronics, and was waiting on the 3d print to finish so I could assemble, but our teammate's 3d printer was faulty and the print failed. Not letting supply chain issues stop me, I grabbed whatever material I had around me and got to work.

Outcome: Maintained 100 % throughput

3.3 Heater-Band & Sensor Integration (Fig. 9, 10)

Specification: 2 × 150 W, Ø38 mm mica bands, 120 V; EPCOS NTC 10 kΩ thermistor.

Mounting: Dual-band stack with 0.13 mm Al foil for thermal spread; routed wires through the phenolic plate

3.4 Manual Pneumatic System (Fig. 3)

The team went to Harbor Freight and called me online so I could guide them on building it: We used all Harbor-Freight fittings and one with a lever, which allowed the simplest and cheapest way to deliver air to both chambers of the piston.

3.4.a. Automatic Pneumatic System (Fig. 4)

5/2 solenoid valve (defective failed coil, 2.1 kΩ → open). 20 mm double-acting cylinder, stroke 150 mm. Delivered remote step-by-step hose routing and startup checks, proved the solenoid had failed.

3.5 Manual Machining & Power Block (Fig. 5)

Machined 1”Dia Steel rod 1” in length: Tapped ½-13 UNC and M10-1.25 half way so we could attach Luke’s motor and the lead screw. This provides a rigid, low-impedance path from the motor to the clamp.

3.5 Manual Machining (Fig.7)

Machined down the plunger we received from david, after we tested the O-ring and found that it would slip off. I took down a few thou to make sure the o-ring could only be taken off by prying it off. This ensured our plunger would never touch the sides of the barrel and create enough pressure for a solid injection.

3. Project Management

Delegated CAD detail to Nico; identified lapse when final drawings lagged. But it was too late to close the design gap before the demo day.

---

4 Analyses Performed

While formal analysis wasn’t part of my assigned responsibilities, I consistently applied hands-on, real-time problem-solving that relied heavily on engineering judgment and practical

diagnostics. I approached issues with a machinist-engineer mindset, especially during high-pressure moments where quick decisions were critical. When our 3D-printed funnel failed mid-run due to printer issues, I quickly analyzed the failure and used available materials, cardboard, duct tape, and the remnants of the failed print to fabricate a functional replacement that preserved 100% throughput. It was a rapid material-flow analysis and a calculated workaround that kept us on schedule.

Another example was when our automatic pneumatic system failed. I diagnosed the solenoid valve issue remotely by guiding hose routing, checking electrical continuity, and ultimately confirming a failed coil through resistance testing. This involved troubleshooting the entire pneumatic circuit under time constraints and identifying the root cause through electrical and mechanical reasoning.

I also carried out precision-fit analysis on the plunger by modifying its diameter to lock the O-ring in place. I measured, tested, and removed just enough material to ensure sealing performance without sidewall contact in the barrel. That decision directly impacted the system's ability to build pressure and inject plastic reliably.

Across all these tasks, my analysis came in the form of quick, informed decision-making backed by hands-on experience and a deep understanding of how the components interacted mechanically, thermally, and pneumatically.

---

5 Conclusion

Hands-on fabrication and rapid troubleshooting were the backbone of my contribution to this project. I kept our injection molder operational through the first demo by constantly adapting to new problems, whether it was wiring errors, machining fitment issues, or last-minute

component failures. Even though we fell short during the second demo, mainly due to design-stage gaps and a failed solenoid, we pushed hard to keep the system functional and made the most out of limited resources.

Looking back, I realize I should've spent more time with the design team earlier in the process. My instinct was to stay in the shop, where I felt most effective, but I've learned that real success comes from tighter collaboration between hands-on builders and designers. If I had taken that extra step, like personally printing the funnel parts or double-checking CAD handoff, we could've avoided some of the breakdowns we hit later. This project taught me that practical experience and design aren't separate skills; they're part of the same workflow. It's the bridge between those two worlds where real engineering happens.

Figures & Tables Referenced in Body

Fig. 1 Polycarbonate Shield



Fig. 2 Emergency-fabricated cardboard funnel.



Fig. 3 Manual Pneumatic.



Fig. 4 Automatic Pneumatic.





Fig. 5 Manually tapping coupler



Fig. 6 Explaining the Process with the team



Fig. 7 Putting the plunger on the Lathe



Fig. 8 Final Product



Fig. 9 Fitting the heating bands on the barrel



Fig. 10 Setting up the electronics

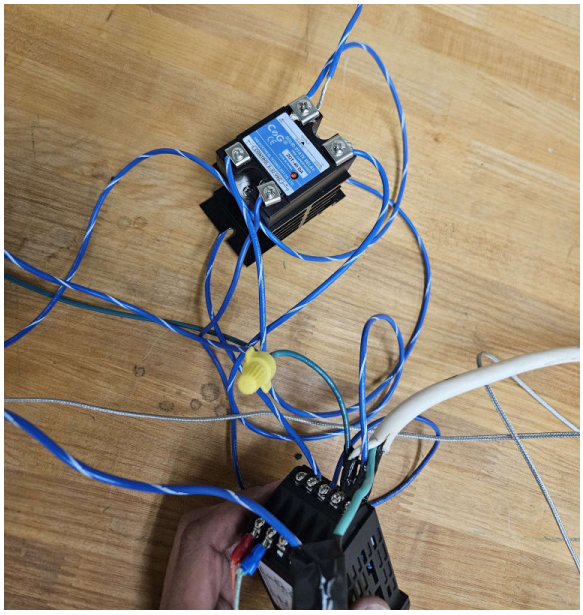


Fig. 11 Building the plastic dispenser

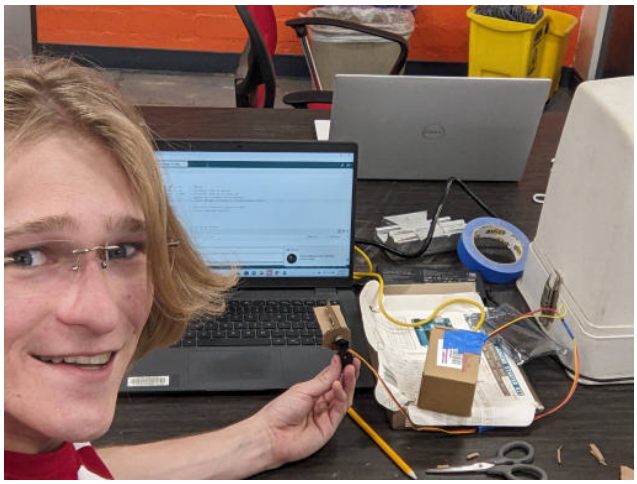


Fig. 12 Testing the dispensing motor

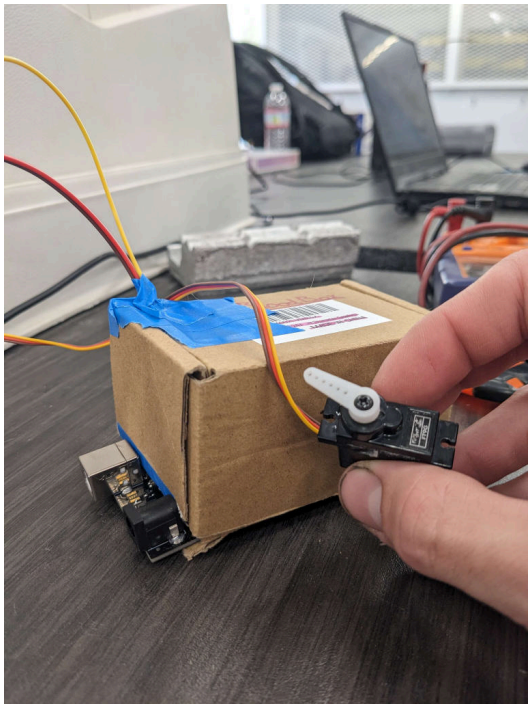


Fig. 13 Built a custom housing box for the electronics

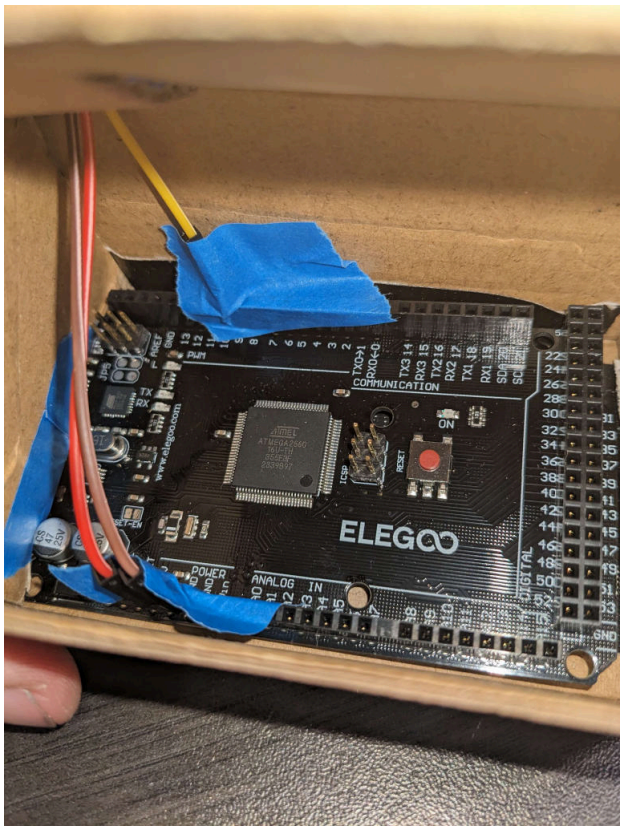


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1. Abstract

The task assigned was to create a vertical injection molding machine from near-scratch under certain design requirements set by our professor John Caffery. For this project, I contributed in many different areas and properly recommended tasks to those who I knew would be a better fit. I had a hand in the design process, fabrication, problem-inquirer, off-campus meets, troubleshooting, acquiring materials and a role in the overall group progress by injecting catalytic creativity to spur brainstorming to solve the myriad of obstacles we had to overcome. I also contributed in safe keeping of the machine and transportation to and from any worksite. One last input I would refer was the motivation I infused when we had to pivot our complete design and we felt like we were falling behind, as a group is only productive when they are in a good headspace and I would hope I injected some of that, along my many other contributions, to help our team persevere.

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2. Introduction

The goal of this project was to create a vertical injection molding machine that is not only adaptable to utilize differing molds, but also be completely automated with minimal human interaction. We were given a time frame of the semester with a CDR, 2 PDRs and one manual test, prior to automation implementation, with a final test being the fully automated

demonstration. Throughout the entire building process, I took part in nearly all corners relating to building a working machine and valued how much I grew from the process, both the successful parts and the troublesome parts and feel more prepared to jump into the industry upon graduation or in an internship.

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3. Design & Contribution

Team Support and Guidance

When it comes to more legislative and management work, I aided the group by reminding deadlines, helping coordinate our many differing scheduling times, securing a place to do much of the physical fabrication of the machine and its components in rancho cucamonga, providing uplift when we felt overwhelmed and went to other sources (i.e. other groups or the Professor for clarification or troubleshooting help), when the group needed such assistance.

I also aided greatly in the intercommunication of the group on any time-sensitive manners and helped us efficiently and considerately resolve any confusions, if any arose. Many forget that being in a team and working on a project isn't just the kind of knowledge or physical labor you bring to the table or finances one can supplement, it also has a large part to do with the attitude and human relations that one introduces into the group as well. One very knowledgeable or talented contributor that has issues working in a team will always be outweighed by any 2 contributors that may not share the same amount of knowledge initially, but will work far more cohesively and positively in a group and therefore yield better results in the end.

Tool Access and Materials

I also provided a fair amount of tools and supplies when it came to physically building the machine and even offered my tools to any other groups if needed. This helped us all feel united in the sense of finishing this educational project. For our team, I utilized the tools I've had for years, space in my car, and the machine itself to be a working transport to allow for any work to be done wherever needed. I also had a large hand in searching for raw components, ensuring capability with the already existing elements upon desired use of the new element proposed and if those parts weren't compatible, attaining a proper substitute or a new plan that would work with what he had on hand at that moment. Many aspects in a project, especially acquiring the hardware/stock, can be fickle and is best to be adaptable to not only preserve the overall timeline, but also avoid any significant design changes that would occur in response as a last resort.

Design Contributions

For the design of this machine, it went through quite many different iterations and while we will not go into each one and why it isn't the final design, it helps to understand the major factors that pushed us from those initial designs and how they led us to where we are now.

From the beginning of the project, we were to design the basic functions of the machine and agree on the overall structure mechanically. In an attempt to reduce the automation complexity, I suggested to our team to eliminate as many “moving” factors to aid in the simplification of the automation by using the injecting force of the cylinder itself to act as the clamping force by using a mold with a seam parallel to the baseplate rather than the orientation that the professor proposed. This eliminates the need to create a whole clamp assembly reducing many complicating factors in the project. However, halfway into our 1st PDR, we came to the understanding that the present injection force and the mold dimensions would overcome the proposed use of the injection force as a clamping force as well as many other complicating elements that would create more issues to solve. This made us pivot to join all other teams in both sections late into the semester. During this time as well, I was a strong advocate for our group in the class session to use circular rods rather than the extrusion rods everyone else wanted to use, as many suggested we switch with them. I was very confident that the rods would hold the forces present and convinced our team to stay steadfast in our decision, which worked in the end. Not only for the costs but also to keep the design simple and ease of attainability of the components. (i.e. ordering the extrusion bars vs a quick trip to a hardware depot.)

While in this more rushed stage, myself and Manny led the charge for a large part in creation of a clamp that can be used within the space of our baseplate, be simple to fabricate and deliver the 3000 pound clamping force. We brainstormed many ideas and brought those ideas to the group until we landed on a cam-shaft style locking mechanism that utilized differing radius sizes and a turning axis. This was a fair idea but upon more analysis and calculations, we found the necessary dimensions to be unusable within our machine, sadly we had to pivot yet again.

At this point, I felt it more necessary to aid the team in the fabrication of the machine itself (as this was prior to the first manual demo) and aid in securing the dimensions of all our components and starting the process of actual fabrication/assembly. Meanwhile, Nico took over the clamp design issue, as he wanted team members that are more acclimated to do the physical assembly work, to work on the machine overall, in which I agreed. As Nico designed and created the innovative clamp, I would check in and offer a 2nd set of eyes and any help, acting as a cohesive team.

Machining and Manufacturing

When it comes to the manufacturing and machining of the injector, I participated a fair amount. I also aided in the machinist drawings to be submitted to the professor, as he graciously and kindly performed a good amount of machining for every group.

I also did a major amount of rough dimensioning and initial cutting of all the members of Nico's clamp design into workable pieces using my friend's bandsaw as well as his drill press in Rancho to perform initial pilot holes for James to further machine later, per figure 1. This was the most critical portion, as we only had the stock on hand and had limited time to utilize the machines there. The plates varied from ½ and inch thick to 1 inch thick aluminum which dictated my cutting speed, increasing the time I had to secure the plate as well as translate the plates themselves forward to be cut, the amount of metal shavings on my body at the end was comical! The pilot holes were just as critical, having one even a relatively small amount off of tolerance would result in an unusable element, further setting us back. After said pilot holes were done, we ran out of time at that location and had to finish at the IO at the highschool in Pomona at another time.

At the IO, I performed machining as well. I used the lathe to perform deeper cuts into the slot where the o-ring fits on the plunger for the cylinder. Nico designed it to be a 'v' shape side profile, however we soon realised that the amount of contact the o-ring needed to stay in position during the whole process wasn't enough. To solve this, I removed the 45 degree cuts and etched out a more deep square profile that allows the o-ring to resist displacement as per figure 7. I also cut the backstop we used for the stationary part of the mold down to a usable height using an angle grinder that fits between the orange center plate and the baseplate. It was galvanized hardened steel but with my diamond coated metal cutting wheel, short work was made of the ⅛ th thick steel per figure 2. Additionally, I used my handheld cutting saw to cut the bottom most wooden base support into the desired dimensions as well as cutting, boring and attaching the wooden 2x4s that support the lead screw element.

One last contribution I had was the day of the final demo, I utilized my people and trade skills to request to David, the lab machinist at CPP, to do a quick clearance hole and square cutout of the aluminum baseplate for the backstop and the ejection exit area of the mold, which was a grand help!

Electrical Work and Motor System Integration

In the electrical and automation area I assisted by doing the complete initial wire connections for the thermo heating bands as well as the PID controller and the relay device per figure 8. This was fun for me as I don't have much experience in wiring and welcomed the challenge to learn. Although my initial wiring wasn't the most efficient wiring for two heating bands, upon the first manual test day the heating chamber was experiencing long heat times and not reaching the desired temp for melting ABS. Upon investigation and help from the professor, we realized I wired it in series when it should be in parallel, preventing full utilization of the heating bands. Accepting I couldn't not figure it out, I requested Hudon's help as I knew he had some knowledge with wiring. This was to prevent our team from falling behind and to not let my

ego bring us down. It was a well earned humbling moment when Hudson finished the wiring and we tested successfully that day. Thanks team!

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4. Analysis

As mentioned prior, this team's machine underwent many different designs and iterations of those designs and in the end landed in a completely different place compared to our initial design. The majority of mathematical analysis that I performed for this project was centered around the initial designs for the "hamburger" mold orientation and the rotating cam idea and was ultimately passed to Nico as I focused on other aspects of the project. However, throughout there was comprehensive communication between all members to ensure that all elements of the design had more than 2 individuals reviewing one member's work, and analysis of the components was no exception. Throughout the remainder of the timeline, I asked Nico many clarifying questions that helped me understand his design and relate any concern, notes and solutions to the group in a timely manner.

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5. Conclusion

In the end, the final machine was not up to par to try a second time and we could not test our final design. Although the design we had at the end of this process was different from the ones in the initial CDR and the final PDR, we are somewhat satisfied with the quick thinking solutions we had to implement as a team when we encountered: unforeseen design error oversights, trouble scheduling meetings to work on the assembly and no consistent location with consistent tooling for long periods. The final product had the elements of the initial design, just not the custom made style to it. We could not fully utilize the clamp design due to spatial interference of the members with each other and the type of motor chosen for final implementation. These solutions would have worked if we as a team collaborated more and had more availability to focus on the project for longer periods of time, we would be as successful on the automation test as the first manual test.

Just knowing the machine we built was nothing but parts before assembly, actually produced a near perfect plastic specimen leads us to be confident in the fact that we would troubleshoot the critical flaws and be successful at automation. Even with this learning error, I feel far more prepared to join any industry in any engineering field on any project and now I have more project experience that will give me a headstart on the construction of my engineering career, so thank you Professor Caffery for the awesome learning experience!

Shoutout to our 3 big campions:
Dereck in Rancho
Martin in the IO
And David in the Lab
Big thank you to you 3 for the help and guidance!

Appendix 7



Figure 1: Here I am rough cutting the 1 inch piece of aluminum using my friends bandsaw in rancho



Figure 2: Here i am cutting down the backstop to height using my angle grinder at the IO



Figure 3: here is our first testing prior to the manual test at the IO



Figure 4: The ejected plastic piece from the first manual test!



Figure 5: The machine at the manual testing

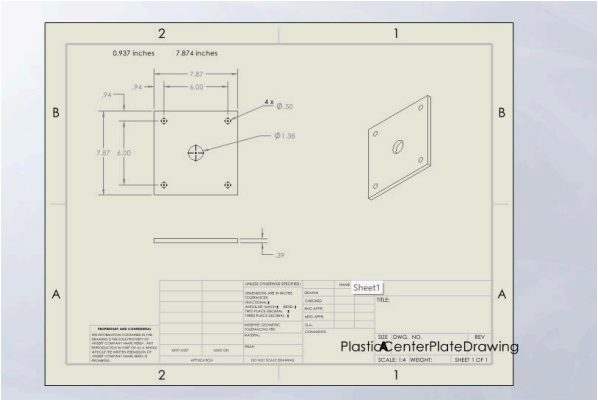


Figure 6: The CAD drawing I did for the center middle plate to be cut by hudson

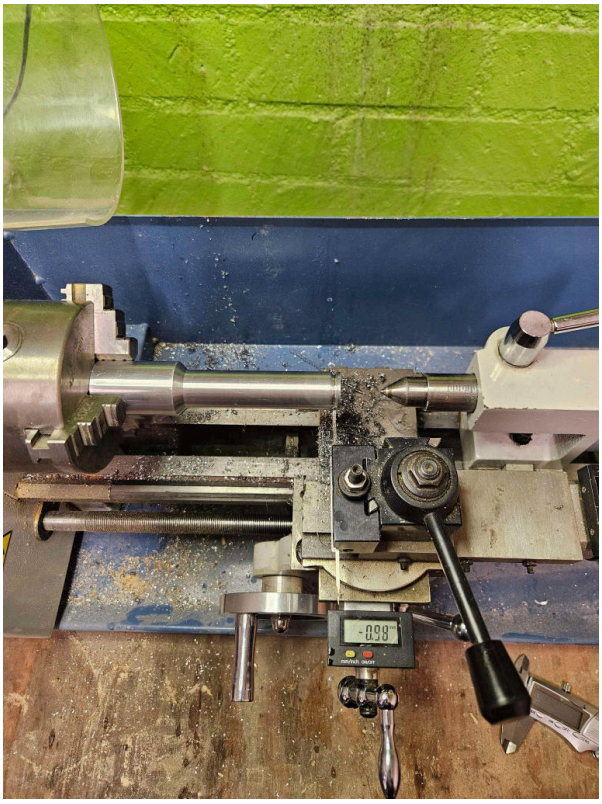


Figure 7: The plunger piece being lathed down for the o-ring to fit snugly that I cut at the IO

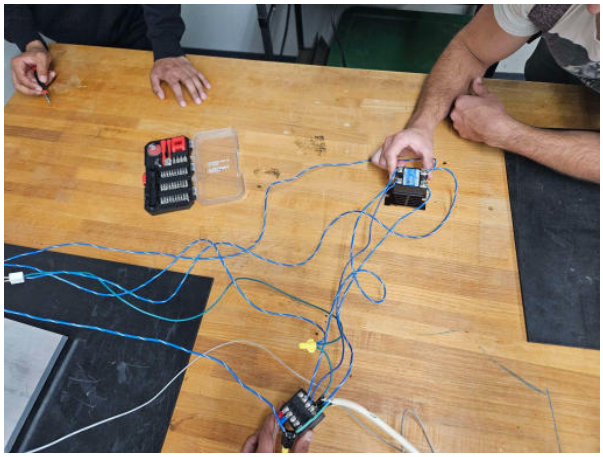


Figure 8: The initial wire set up I started for the first manual testing



Figure 9: Here is the final product, we had our ups and downs but I'm still proud that we had something to show for it along with a lot of knowledge and experience gained!